

UNIVERSITÄT
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Tutorial 3

Basic Programming Paradigms

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Outline

- MC Questions
- Exercise 1 – Tracing
- Programming
- Questions

Question 1: How many values does a function with the return type void return?

- exactly one
- none
- two
- three or more

Question 2: Which of the following is NOT a valid control structure in C++?

- For-loop
- If-statement
- Repeat-loop
- While-loop

Question 3: What is the role of the main() function in a C++ program?

- To provide memory allocation for the program.
- To serve as the entry point for program execution.
- To compile the program.
- To define global variables used across the program.

Question 4: What is true for the following lines of code?

```
if(a>b){
    int temp = a;
    a = b;
    b = temp;
}
```

- The code uses a loop to iterate over the variables a, b and temp.
- If b is greater than a, the code fragment performs a swap of both variables.
- The variable temp can store floating point values.
- If a is smaller than b, the code fragment does not get executed.

Question 5: Consider the following code fragment. Which statement is wrong?

```
int a;
```

- A variables with the name "number" gets declared and defined.
- The value of number is undefined at this point.
- The variable number can be safely used from now on.
- No initialization is performed.

Question 6: Which output stream is NOT buffered?

- clog
- cout
- cin
- cerr

Question 7: What is the correct syntax to access an element from the namespace std?

- std::cout
- cout::std
- std->cout
- std.cout

Question 8: Your main function returns the value 1. Which implications does this have?

- I can use the return value of the function to perform further calculations.
- The main function was successfully executed.
- The program crashed during execution.
- An error occured during execution.

Exercise 1 - Tracing

Exercise (20 minutes)

- Solve exercises 1 to 3.

Programming

Sequence of numbers (10 minutes)

- Write a program, which outputs the following sequence of numbers: 0, 1, 3, 6, 10, 15, … (<100)
- Write three different functions, which use different kinds of loops to calculate the sequence. All three functions should produce the same output.
- Compare each function and decide which loop is the most appropriate for solving this task.

ASCII table

- Maps characters to numerical (integer) values, which can then be represented as binary values

e.g.:

Decimal	Hex	Char
48	30	‘0‘
49	31	‘1‘
...	...	..
65	41	‘A‘
66	42	‘B‘
...	...	...
97	61	‘a‘

ASCII

- Write a function which outputs every character of the ASCII table (0 – 127).

ASCII dec.	character
...	...
65	A
66	B
...	...

ToUpper() (5 minutes)

- Write a function that takes a string and converts all lowercase letters to uppercase letters.

Notes

- Use the following signature:

  string toUpper(string input)

  { //function body }

- Use *input.at(i)* with *i* as a zero-based index to access the individual characters of the string.
- Use *input.size()* to determine the size of the string.

Questions?

Feedback



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THANK YOU!

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